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Roll No

AU-7005(1)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Renewable Energy

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Write down the difference between the extra-terrestrial and terrestrial radiation.
b) List the instruments used for measuring solar radiations? Discuss any one.
2. a) Explain the working of solar water heater with a neat sketch as an application of solar thermal systems.
b) Explain the performance of PV cell with a neat equivalent circuit diagram.
3. a) Classify the WECS.
b) Write short notes.
 - i) Power torque and speed characteristics of wind turbine.
 - ii) Application of wind turbine.
4. a) Explain in detail about the configuration of horizontal and vertical axis wind.
b) Give the classification of biomass. How energy from biomass can be maximized?

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5. a) What is Biomass?
b) Explain anaerobic digestion.
6. a) What are small hydropower plants and how do you classify them?
b) Explain the design criterion of turbines in case of small hydro system.
7. a) What is geothermal energy?
b) What are advantages of Hydrogen as a fuel?
8. Discuss any Three
 - i) Solar dryers
 - ii) Principle of ocean wave energy conversion.
 - iii) Differentiate between C3 and C4 plants with respect to photosynthesis
 - iv) Hydrogen Energy.
